

LOKAVERKEFNI

KEST3NL05EU

Weight: 20%

Start Date: 29-Apríl-2026

Due Date: 20-Maí-2026



UPPLÝSINGATÆKNISKÓLINN

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Linux - netstjórnun

Objectives:

The objective of this project is to build a reliable, secure, and centralized Linux-based server infrastructure for DDP ehf. that provides essential network services to 30 employees across four departments. The infrastructure will support IP management, name resolution, user authentication, secure remote access, centralized logging, email services, printing, time synchronization, and automated backups.

Scope

- **Server:** server1.ddp.is (Central Management Server)
- **Clients:** client1.ddp.is (Debian) and client2.ddp.is (Red Hat)
- **Domain:** ddp.is
- **Network Architecture:** Dual network interface configuration on server1

Network Architecture

Server1 will be equipped with **two network adapters** as shown in the following table

Adapter	Type	Purpose	IP Address/Configuration
ens33	NAT (VMnet8)	Internet access (for updates & packages)	Dynamic (DHCP from VirtualBox/VMware NAT)
Ens34	Host-Only (VMnet1)	Private company network	Static – 192.168.100.10/24

- The **private interface (ens34)** will host all internal services (DHCP, DNS, NTP, Syslog, CUPS, SSH, etc.).
- The **NAT interface (ens33)** will be used only for internet connectivity to download packages and updates.
- All clients (client1 and client2) will be connected exclusively to the **192.168.100.0/24** private network.

IP Addressing Plan:

- **server1.ddp.is** → 192.168.100.10 (10th usable IP on the private network)
- Default Gateway on private network: 192.168.100.1 (or leave as none if no external routing is required inside the lab)
- DNS Server: 192.168.100.10

Project Instructions:

1. Install and configure **server1**, **client1**, and **client2** with hostnames and domain ddp.is.
2. Configure **server1** with **two network adapters**:
 - One NAT adapter for internet connectivity.
 - One private adapter with static IP address **192.168.100.10/24** from the 192.168.100.0/24 block (10th usable IP).
3. Install and configure **DHCP server** on server1 to automatically assign IP addresses, gateway, DNS, and domain name to client1 and client2 via the private interface.
4. Install and configure **DNS (BIND9)** on server1 for forward and reverse name resolution within the ddp.is domain.
5. Create all 30 user accounts using an automated script based on the provided **Users file**.
6. Schedule weekly backups of all home directories to run every **Friday at midnight**.
7. Install and configure **NTP** with server1 as the master time server synchronizing client1 and client2.
8. Configure **server 1** as a centralized **Syslog** server to collect logs from both clients for proactive monitoring.
9. Install and configure **Postfix** mail server on server1 integrated with **Roundcube** webmail for

- sending and receiving emails.
10. Install and configure shared printers using CUPS with group-based access control. Only users belonging to their respective department groups may print. **IT** and **Management** groups shall have both print and manage rights.
 11. Harden SSH on all systems: Disable password authentication and enforce **RSA key-based authentication** only.
 12. Close all unused ports on server1 and clients. Verify the firewall configuration using **Nmap** scans.

Note: All configuration files must be well-commented. Screenshots should clearly show service status (systemctl status), configuration files, successful DHCP lease, DNS resolution, SSH key login, backup execution, and Nmap results. The final submission must be uploaded to GitHub before the due date and include the following in a clean, professional structure:

```
/DDP-Linux-Infrastructure-Project/
├── README.md          (Executive Summary + Project Overview)
├── Documentation/
│   ├── Project_Report.pdf
│   ├── Screenshots/
│   ├── Network_Diagram.png
│   └── Configuration_Guide.md
├── Scripts/
│   ├── create_users.sh
│   ├── backup_home.sh
│   └── system_hardening.sh
├── Config_Files/
│   ├── dhcpd.conf
│   ├── named.conf
│   ├── named.ddp.is.zone
│   ├── sshd_config
│   ├── rsyslog.conf
│   ├── postfix_main.cf
│   └── ntp.conf
├── Evidence/
│   ├── nmap_scans/
│   ├── user_list_verification.png
│   └── service_status_screenshots/
└── LICENSE
```

Teacher
Abdelaziz Ghazal (AGH)
agh@tskoli.is